Fiscal Year:	FY 2009	**Task Last Updated:**	FY 09/28/2009
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	NSCOR: Solid Tumor Risk Estimation: Incorporating Intercellular Interaction Effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02135-2907	**Congressional District:**	8
Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSCOR Space Radiation NNH04ZUU002N
Start Date:	10/01/2004	**End Date:**	10/31/2010
No. of Post Docs:	10	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	12
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date should be 10/31/2010, per M. Klumpar in PI's office [Ed., 3/28/2011] NOTE: Received NCE to 3/31/2011 per JC. Guidry/JSC (10/2010) NOTE: Received NCE to 9/30/2010 per J. Dardano/JSC (9/09) NOTE: New grant number NNJ06HA28G received in 2006 when PI moved to Tufts and end date is 9/30/2009 (information received 6/2009) jvp NOTE: End date changed back to 9/30/2009 per S. Krenek/JSC (8/07) End date changed to 10/31/2009 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:	Dr J. Folkmann deceased		

COI Name (Institution):	Folkman, Judah M.D. (Children's Hospital, Harvard Medical School) Hahnfeldt, Philip Ph.D. (Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine) Huber, Peter (German Cancer Research Center (dkfz)) Lamont, Clare (Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine) Sachs, Rainer Ph.D. (University of California)
Grant/Contract No.:	NNJ06HA28G ; NNJ04HJ12G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	We will improve solid tumor risk estimation for astronauts. Improvements will be based in large part on analyzing data resulting from our high-energy iron ion and proton in vivo irradiations. Our mathematical risk modeling will use standard computational approaches which minimize theoretical assumptions and also use minimally-parameterized, biologically-based carcinogenesis models capable of interrelating epidemiological data, animal experiments, and in vitro radiobiology. In our experiments and our computational analyses, special emphasis will be placed on tumor promotion and progression, as influenced by intercellular signaling among various cell populations within the tumor (e.g. tumor cells, endothelial cells and stroma). The in vivo models used are highly suitable for the program. These include the transgenic conditional k-ras lung tumor model from the Tyler Jacks's laboratory that will allow the study of radiation modulation of all steps in carcinogenesis. These mice have a recombinant adenovirus expressing Cre recombinase to induce k-ras expression which synchronizes tumor onset, allowing numerous tumors to develop simultaneously in the lungs. The number of tumors and rate as which they progress though the four stages of carcinogenesis (e.g. time-dependent ratio of hyperplastic to neoplastic lesions) can then be traced as a function of the timing, dose and quality of the radiation. We will also use dormant human breast tumor xenograft models which allow us to place particular emphasis on dissecting the progression stage of carcinogenesis. These in vivo models offer specific advantages: solid tumor sites of concern to NASA (lung, breast, and later thyroid) will be studied; middle-age mice will be used; and due to the nature of these models comparatively few mice will be needed and post-irradiation study times will be comparatively short. In this Program we implement a multi-level approach with whole organism-, tissue- cell-, and molecular-level endpoints used to measure radiation response. For examples, in addition to radiation influence on latency periods or the time-dependent ratio of dysplastic versus frankly neoplastic lung lesions, we examine cell signaling, use matrix and clustering computer algorithms for analyzing transcriptome data, and interpret chromosome aberrations scored with mFISH/SKY using our established computer simulation software. Importantly for our emphasis on intercellular signaling as a key aspect of carcinogenesis, we will assay not only tumor cells but also tumor-associated stromal and endothelial cells (including circulating endothelial cells). It is turning out that such tissues can help support or repress tumors (i.e. play a major role in promotion and progression), and due to the relative stability of their genomes they are simpler to analyze for radiation-induced DNA damage. An experimental and theoretical emphasis on tumor progression is also planned, based on the fact that this step in carcinogenesis has hitherto received less attention from radiation risk modelers than other steps, e.g. initiation, but is at least as important. There is now strong evidence that microscopic dormant neoplastic sites are far more prevalent in adults than previously assumed and that their progression can be accelerated by radiation. Radiation shortening of latency periods could thus be a key component of solid tumor risk for middle-aged astronauts. A tightly-knit interdisciplinary team, Director L. Hlatky, Associate Director R. Sachs along with the other Project leaders J. Folkman, P. Huber, and P. Hahnfeldt will carry out the 5 closely interrelated Projects: (1) mouse models for assessing carcinogenesis risk; (2) HZE and low LET irradiation; (3) radiation transcriptome analyses; (4) quantitative chromosome aberration analysis; and (5) quantitative estimation of solid tumor risk. Project (5) will integrate data from the first four Projects as well as drawing on results from the literature. It will culminate in a composite radiation carcinogenesis model designed to reduce the uncertainties in risk estimates for astronauts by addressing cell interaction effects and including in the multi-step carcinogenesis analysis the overlooked step of progression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Autopsy findings from adults dying from non-cancer causes have now unequivocally established that most of us possess fully-malignant tumors that are held in check by a balance between cell cycling and cell death. Such tumors are characterized by a failure to advance through the angiogenic switch. Dormant tumors that manage to switch to the angiogenic phenotype grow exponentially and ultimately become clinically apparent, which is referred to as the "disease state" of cancer (Folkman, Nature 2004). As recent results from Project 2 disclose, major angiogenic genes show significant regulation by space radiations. This suggests there may be a link between space radiation exposure and modulation of the dormancy bottleneck that holds existing indolent lesions in check. To research the matter, the development of dormant tumor models has been a priority. We now have in place several human dormancy models in mice. The models have been phenotypically characterized (Almog and Folkman, JNCI 2006 and FASEB 2004). Tumors remain dormant for a prolonged period of time until they spontaneously switch to the fast-growing phenotype. The time point of the switch is tumor-type dependent. We recovered several of the spontaneously-switched tumors from mice and cultured them in vitro. Re-implantation in naive mice confirmed a stable fast-growing phenotype. It is concluded a genetic reprogramming event accompanies transition from dormancy to fast-growth. We have designed new methods for estimating the incidence of iatric (i.e. treatment induced) cancers after radiotherapy. As cancer radiotherapy methods improve and the number of patients treated increases, more and more people survive long enough after the initial cancer diagnosis to potentially develop radiotherapy-induced second malignancies. Because treatment techniques are changing rapidly, there is insufficient follow-up time to directly measure the second cancer risks associated with the newer methods. Biologically-based mathematical models, which implement plausible assumptions about radiation carcinogenesis processes such as initiation of organ-specific stem cells to the pre-malignant state, inactivation (killing), and repopulation of both normal and pre-malignant stem cells during radiotherapy and subsequent organ recovery, can address this problem: Models calibrated with older data can predict second cancer risks of new/prospective radiotherapy protocols.

a) Estimating spatial range and relaxation times for radiation bystander effects, and their implication for responses to GCR fields. Using a "fire, diffuse, fire" signaling model, we analyzed an experiment of Belyakov et al. where only a thin plane of cells in a 3-D artificial skin system was irradiated with a microbeam, and bystander effect endpoints (apoptosis and micronuclei) were measured in parallel planes located at various distances away from the irradiated plane. Adjusting the model parameters to the data indicates: A single cell hit by alpha particles can influence ~100-2000 bystander cells. Characteristic relaxation times could be on the order of weeks or even longer. We applied these estimates to the question of mixed GCR fields, using criteria for additivity and linearity of multi-track action obtained from analyzing PFGE experiments. We found that if the estimates obtained for this in vitro tissue-mimetic system and for two surrogate endpoints apply to radiation carcinogenesis in vivo, multitrack additivity cannot be counted on at GCR flux levels. There may be additivity, synergy, or agonism, but the flux levels where additivity can theoretically be inferred from independence of effects of different tracks are, due to the large target size and long relaxation time, lower than that found on a space voyage.
b) Introduced population-level growth limitations to the promotion of initiated cells. We developed a Two-Stage Logistic model as a deterministic alternative to the widely-used TSCE model that begins to account for population-level growth inhibition on the advancement of initiated cells. It explicitly considers a density-dependent slowdown in the accumulation of initiated cells.

It is now well appreciated that cell-cell interactions can modulate the advancement of cells through the steps in carcinogenesis, yet modeling has been slow to incorporate these interactions. Indeed, we now know that nascent tumors can become dormant, an example of how inhibitory population-level effects can curtail innate cell-level expansion. The TSL model focuses specifically on this cell-population interface during promotion. Although deterministic, the model can provide reasonable agreement with the atom bomb data. It can do this because it handles the short-term radiation effect in a similar manner independent of its tendency to initiate or to promote. The TSL model is minimally parameterized, which is important when using epidemiologic data to back-infer parameter values.

c) Considered the influence of cell-cell interactions on the role of basic cell kinetics on tumor promotion and progression in silico, and found complicated risk implications.

It was found, just by considering the combined kinetics of tumor cell proliferation capacity, migration and cell death in a realistic cell-cell interaction setting, unexpected U-shaped dependencies of tumor growth on low-dose radiogenic cell killing are observed. When the migration rate is small, a single cancer stem cell can only generate a small, self-limited clone due to the finite life span of progeny and spatial constraints. The tumor, although present, never rises to clinical incidence, and the risk of tumor occurrence is zero.

By contrast, a high migration rate can break this equilibrium, seeding new clones at sites outside the expanse of older clones. In this manner, the tumor continually “self-metastasizes”. Counterintuitively, when the proliferation capacity is low and the rate of cell death is high, tumor growth is accelerated due to the freeing up of space for self-metastatic expansion. We conclude that changes to proliferation and cell death that increases the rate at which cells migrate can cause a tumor to transit from a subclinical clone to manifest cancer disease. Given the prevalence of dormant clones in the average adult, a significant possibility exists that space radiations may alter cancer risks through such a mechanism.

d) Explored mathematically the role of the tumor/microenvironment interaction in vivo in enabling tumor progression from a zero-risk dormant state into a higher-risk disease state. It was found using an animal model of angiogenesis-dependent tumor growth that an important mechanism for the maintenance of the dormant state is angiogenesis inhibition. In terms of the timeline paradigm, this emphasizes the critical role progression-level tissue-interactions play in carrying a tumor to clinical incidence. Accordingly, it is paramount that an existing angiogenic balance not be tilted towards stimulation. We have shown in Project 2 that iron irradiations may cause such a shift, and that protons might do the opposite. A mathematical construct has been refined that captures the epithelial/endothelial dynamic controlling the balance mechanism. It recasts the population biology concept of a fixed “carrying capacity” for the growth of organisms in stable environments to account for the ability of a tumor to regulate its own angiogenic carrying capacity. This “dynamic carrying capacity” model captures an important, final “bottleneck” event standing in the way of tumor incidence. By coupling developments here with the quantitative cell-level and promotion-level modeling we are developing in parallel using mouse and tissue culture platforms, the goal is to produce a comprehensive, yet practical, model for radiogenic cancer risk.

e) The prospect that cell-cell fusion may modulate carcinogenesis progression introduces a new set of mathematical considerations in radiogenic cancer risk, including the role of chromosomal instability.

Lacking currently is a satisfactory explanation for how cells displaying chromosome instability and aneuploidy persist given the generally deleterious nature of these mutations. Using evolutionary theory, we show quantitatively that while the diversity accompanying aneuploidization is insufficient to overcome losses due to deleterious mutations, a newly discovered ability of tumor cells to fuse with host cells might provide the extra genetic information necessary to explain their advancing fitness.

Our group has already shown there is a naturally-occurring fused cell fraction in mixed tumor/normal cell populations (specifically, mixtures of murine lung fibroblasts and Lewis lung cancer cells). The addition of extra non-mutated gene copies in the fused cells may allow these cells to overcome the deleterious effects of mutations. It is concluded that a fusion mechanism may play a role in sustaining progression-phase carcinogenesis in the face of genomic instability.

f) Age at time of exposure to Fe irradiations was found to influence cancer risk through alterations of interactions between tumor cells and non-transformed host cells. It is established that tumor growth, and therefore incidence risk, is highly dependent on stromal permissiveness. To begin to understand how the age of the individual influences tissue permissiveness and risk of cancer incidence due to space irradiation, we assayed tumor incidence of mice injected with human lung tumor cells A549 following fractionated, low-dose Fe (1GeV) radiations (0.2Gy x 5). Young and old mice were found to differ significantly in their ability support of the growths of injected A549 human lung tumor cells. The young mice showed a more rapid tumor onset and 100% developed tumors and died as a result of there tumors. This was compared with only 60% of the older mice who developed tumor burden.

Task Progress:

g) Tumor cells were found to interact with surrounding stromal cells, even at great distances, to modify their own microenvironment. While the ‘bystander effect’ has been well studied in the context of a radiation damage response that

can propagate from targeted to non-targeted cells over large ranges, the possibility this might occur spontaneously in the tumor setting has not been considered. Here we show in vivo and in vitro that cancer cells constitutively display DSB damage, and that stromal cells in proximity to these cancer cells display increased amounts of DNA/repair foci.

Cancer cells exhibit significant constitutive levels of DNA breakage and repair, detectable via g-H2AX and 53BP1. This constitutive breakage/repair may indicate the continuous generation and processing of damage in tumor cells and may provide insight into important mechanisms driving genomic instability. Cancer cells also induce DNA DSB/repair foci in stromal cells. In vivo studies indicate DSB/repair foci detectable by g-H2AX and 53BP1 expression increases for stromal tissues in the immediate tumor microenvironment. It is concluded that the radiogenic bystander effect may be an amplification by radiation of a stress communication that is already constitutively active. This suggests an altered paradigm for how radiation couples to oncogenic stress signaling throughout the tumor population. More broadly, tumor cells themselves are found to play an important part in conditioning the tumor microenvironmental niche to tumor development, and therefore cancer incidence risk.

h) In addition to angiogenesis reduction, we demonstrated that proton irradiations may reduce cancer incidence risk by reducing proliferation, invasion and tumor growth rate during the progression phase. When a Matrigel-coated semi-permeable membrane was placed over media containing serum (lower chamber) and proton-irradiated tumor (A549), fibroblast, or endothelial cells were placed atop the membrane (upper chamber), a reduction in cell invasiveness towards the media compartment was noted. The results extend on the finding that angiogenic factor production is reduced by protons. Reductions were about 47% for tumor cells and about 12% for endothelial cells ($p < 0.05$).

Reversing the above, when proton-irradiated fibroblasts were placed in the lower chamber and endothelial cells were placed atop the membrane in the upper, endothelial invasion was reduced over that noted when the fibroblasts in the lower chamber were not previously irradiated. Results were analogous for the proliferation assay. In this case, when endothelial cells were placed in the lower chamber and proton- or gamma-irradiated tumor or fibroblast cells were placed in the upper, endothelial cell proliferation in the lower chamber was reduced for proton irradiation and heightened for gamma. These results accord with our other data showing that protons decrease pro-survival angiogenic factors, while gamma irradiations enhance their expression. Finally, growth of proton-irradiated A549 tumor cells implanted in nude mice exhibited slowed tumor growth after correcting for cell death.

These results suggest that proton radiations may reduce symptomatic cancer risk through favorable reductions in a number of progression-level behaviors that define tumor development and metastasis.

i) We have designed animal models to explore the molecular underpinnings of the “angiogenic switch”, a master controller of tumor dormancy that can modulate final radiogenic cancer risk. Autopsy findings from adults dying from non-cancer causes have now unequivocally established that most of us possess fully-malignant tumors that are held in check by a balance between cell cycling and cell death. Such tumors are characterized by a failure to advance through the angiogenic switch. Dormant tumors that manage to switch to the angiogenic phenotype grow exponentially and ultimately become clinically apparent, which is referred to as the “disease state” of cancer (Folkman, Nature 2004).

As recent results from Project 2 disclose, major angiogenic genes show significant regulation by space radiations. This suggests there may be a link between space radiation exposure and modulation of the dormancy bottleneck that holds existing indolent lesions in check. To research the matter, the development of dormant tumor models has been a priority. We now have in place several human dormancy models in mice. The models have been phenotypically characterized (Almog and Folkman, JNCI 2006 and FASEB 2004). Tumors remain dormant for a prolonged period of time until they spontaneously switch to the fast-growing phenotype. The time point of the switch is tumor-type dependent. We recovered several of the spontaneously-switched tumors from mice and cultured them in vitro. Re-implantation in naive mice confirmed a stable fast-growing phenotype. It is concluded a genetic reprogramming event accompanies transition from dormancy to fast-growth.

j) Using the dormancy models, we elucidated potential angiogenic switch genes regulated by space radiations. Using endostatin as an angiogenic gene network probe, we uncovered an extensive consensus transcriptional “program” underling the conversion of the four dormant tumors to the fast-growing angiogenic phenotype (Abdollahi and Folkman Mol Cell 2004, PNAS 2007). We applied a similar methodology to dissect the molecular mechanisms of tumor dormancy by characterizing the consensus transcriptome signature of dormant vs. fast-growing experimental human osteosarcoma, liposarcoma, breast carcinoma and glioblastoma multiforme tumor models in immunocompromised SCID mice.

In agreement with our phenotypic observation, the angiogenesis-related genes were the most significantly enriched functional category of genes distinguishing dormant vs. fast-growing tumors. We have shown differential regulation of mRNAs and proteins underlying the phenotypic switch of dormant to fast-growing tumors. Several of these genes and pathways are related to the angiogenesis process (e.g. thrombospondin and angiomotin) but we also found genes that were not previously linked to angiogenesis process, such as Histone cluster 1 protein H2BK that was upregulated in all dormant tumors. We found that a key feature of the switch of dormant tumors to angiogenic fast growing tumors was the loss of antiangiogenic stimuli, but activation of EGFR, IGF1R and PI3K signaling also occurred. Together, these data provide a better understanding of the molecular mechanisms underlying tumor dormancy and the switch of dormant tumors to fast-growing phenotype.

k) Genome-wide transcriptome analysis reveals that proton radiation downregulates key pro-angiogenic proteins, including VEGF, HIF-1alpha and IL8. This seems to be a unique property of proton irradiation as compared with iron or carbon ions. In alignment with this observation we found reduced capability of human A549 lung tumor cells to recruit blood vessel and to colonize the lungs of Balbc nude mice. We further found that microvascular endothelial cells are more sensitive to proton radiation compared to tumor cells or stroma cells (fibroblasts). Although proton irradiation shares some physical features with heavier ions such as carbon (both demonstrate a Bragg peak), the classical biophysical analysis of protons has found no substantial difference in biologic activity between protons and photons.

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